



**ALZ-Foresight: Predicting Alzheimer's Disease
Progression Modeling Using Multimodal Machine
Learning on the ADNI Dataset**

by

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Abstract

Alzheimer’s disease is a progressive neurodegenerative disorder that affects memory, cognition, behavior, and independent daily functioning. Early detection and reliable stage classification are important because clinical interventions, care planning, and research monitoring depend on timely and interpretable information. This final report presents the full lifecycle of *ALZ-Foresight*, a multimodal machine learning and web-based decision-support project developed using the Alzheimer’s Disease Neuroimaging Initiative (ADNI) dataset.

The project integrates a leakage-safe structured ADNI branch, a CNN-based MRI image branch, a late-fusion multimodal prediction branch, and a deployed web application for prediction, visualization, and explanation. The structured branch merges six ADNI tables at visit level, applies subject-level splitting using RID before preprocessing, filters high-missingness variables using training-only statistics, and produces 111 processed features. XGBoost was found to be the strongest structured model. The tuned Top-15 XGBoost model achieved 91.90% test accuracy, 0.9189 weighted F1, and 0.9783 weighted one-vs-rest ROC-AUC. The CNN image branch achieved 70.47% holdout accuracy and 0.8506 ROC-AUC, while the multimodal late-fusion model achieved 73.15% accuracy and 0.8254 ROC-AUC.

The implementation includes a React web interface, Node.js/Express backend, MongoDB Atlas database, Python model inference layer, risk visualization, AI-generated insights, audit logging, model management, and security controls. The system passed 19 out of 19 functional and non-functional test cases, covering authentication, role-based access control, patient management, input validation, prediction inference, visualization, explainability, logging, model versioning, security, and reliability. The project remains research and educational software, not a medical device. Future work includes stronger MRI preprocessing, improved CNN training, calibrated multimodal fusion, external validation, proxy-feature ablation, and explicit longitudinal progression modeling.

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The team would like to express sincere gratitude to Dr. Mohamed Tharwat for his continuous supervision, guidance, and feedback throughout the development of this graduation project. His academic and technical direction helped shape the project from proposal to implementation, testing, and final documentation.

Data used in this project were obtained from the Alzheimer’s Disease Neuroimaging Initiative (ADNI) database (adni.loni.usc.edu). The ADNI was launched in 2003 as a public-private partnership led by Principal Investigator Michael W. Weiner, MD. The primary goal of ADNI has been to test whether serial magnetic resonance imaging (MRI), positron emission tomography (PET), biological markers, and clinical and neuropsychological assessment can be combined to measure the progression of mild cognitive impairment and early Alzheimer’s disease.

Data collection and sharing for this project were funded by the Alzheimer’s Disease Neuroimaging Initiative through the National Institutes of Health Grant U19AG024904. This project uses ADNI data only for academic and research purposes and follows the principles of de-identification, non-redistribution, and responsible use of research data.

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Chapter 1

Introduction and Project Proposal

1.1 Introduction

Alzheimer’s disease is one of the most important neurodegenerative conditions affecting aging populations. It causes progressive deterioration in memory, cognition, behavior, and independence. The disease does not appear suddenly; rather, it often develops through a gradual continuum that includes cognitively normal status, mild cognitive impairment, and Alzheimer’s disease dementia. For this reason, computational systems that can combine clinical, cognitive, genetic, and imaging data may provide useful support for research and early decision-making.

The *ALZ-Foresight* project was proposed as a multimodal machine learning and web-based decision-support system that uses ADNI data to classify Alzheimer-related disease stages and support progression-aware analysis. The project combines machine learning methodology with software engineering implementation. It includes data preparation, multimodal model development, system requirements specification, software design, web implementation, testing, evaluation, and reporting.

1.2 Problem Definition

Existing tools for Alzheimer’s research and prediction are often fragmented. A researcher or clinician may need one tool for cognitive scores, another for imaging data, another for prediction, and another for visualization. This fragmentation creates several problems. First, it makes it difficult to combine different modalities such as MRI images, cognitive scores, diagnostic labels, demographic variables, and genetic information. Second, it makes it harder to ensure that data leakage is prevented in longitudinal datasets, especially when several visits from the same subject are available. Third, it limits in-

interpretability because many machine learning outputs are presented as isolated numbers rather than clinically understandable results.

The formal problem addressed by this project is therefore:

How can a leakage-safe, multimodal, and reproducible machine learning system be designed and implemented to classify Alzheimer’s disease stages from ADNI data while also providing usable web-based prediction, visualization, explainability, and system validation?

1.3 Motivation

The project was motivated by both medical and technical needs. From the medical side, Alzheimer’s disease requires earlier and clearer detection support. Researchers and clinicians benefit from systems that can organize patient information, summarize risk, and show longitudinal clinical context. From the technical side, Alzheimer prediction is a challenging machine learning problem because ADNI data contain multiple modalities, missing values, repeated visits, class imbalance, and large imaging files. These factors make the project suitable for applying data engineering, machine learning, deep learning, software architecture, database design, API integration, and testing.

The project also has educational value because it demonstrates the full development lifecycle of a real applied AI system. It starts from proposal and requirements, continues into system design, proceeds through implementation, and ends with testing, evaluation, and reflection.

1.4 Project Objectives

The main objectives of ALZ-Foresight are:

1. Build a reproducible ADNI preprocessing pipeline using subject-level splitting to reduce leakage risk.
2. Train structured machine learning models using clinical, diagnostic, demographic, cognitive, and genetic features.
3. Develop a CNN-based image branch for MRI-based Alzheimer stage classification.
4. Implement a late-fusion multimodal approach that combines structured and image-based predictions.

5. Build a secure web application for user authentication, patient management, prediction inference, visualization, and explanation.
6. Validate the system using functional and non-functional test cases.
7. Document the project using the ESLSCA graduation thesis structure.

1.5 Project Scope

The project scope includes ADNI data ingestion, structured data preprocessing, CNN image modeling, late-fusion prediction, model evaluation, web application development, API integration, patient management, role-based authentication, model versioning, risk visualization, explainability panels, logging, security testing, and reliability evaluation.

The project is research and educational software only. It is not intended to function as a certified medical device and does not provide clinical diagnosis by itself. It does not include direct hospital EHR integration, regulatory approval, bedside deployment, or real patient acquisition outside permitted ADNI use.

1.6 Proposed System Overview

The proposed system accepts structured clinical data and MRI-related image inputs. The structured branch processes cognitive, diagnostic, demographic, and genetic ADNI variables. The image branch processes MRI samples through a CNN classifier. The fusion branch combines model outputs using a late-fusion strategy. The web application provides a dashboard where authorized users can manage patients, enter assessment values, request predictions, view class probabilities, and read AI-generated explanations.

1.7 Project Deliverables

The final project deliverables include:

- Project proposal and proof-of-concept.
- Software Requirements Specification.
- Software Design Document.
- Machine learning notebooks and trained model outputs.
- CNN and fusion experiment outputs.

- Web application source code.
- Test cases report.
- Research paper.
- Final graduation report.

1.8 Project Timeline

Table 1.1: Project phases and timeline

Phase	Start	End	Main Tasks
1	20/09/2025	18/10/2025	Proposal and proof-of-concept, including early baseline and demo flow.
2	18/10/2025	15/11/2025	SRS document and initial code stubs for front-end, backend, and ML service.
3	15/11/2025	18/12/2025	System design, architecture, database schema, UI wireframes, and core endpoints.
4	18/12/2025	04/04/2026	Implementation, research paper, model development, and preliminary results.
5	04/04/2026	25/04/2026	Testing, evaluation, security checks, reliability checks, and bug fixing.
6	25/04/2026	13/05/2026	Final thesis report, documentation consolidation, and defense preparation.

Chapter 2

Background and Literature Review

2.1 Background

Alzheimer’s disease is commonly studied through multiple data types. Clinical and cognitive assessments such as MMSE, CDRSB, ADAS-Cog, FAQ, and mPACC provide structured measures of cognitive and functional status. MRI images provide anatomical information related to brain atrophy and structural change. Demographic and genetic variables, such as age, gender, education, and APOE-related information, can contribute additional risk context.

The ADNI dataset is widely used in Alzheimer’s research because it provides longitudinal clinical and imaging data collected across several phases. However, using ADNI correctly requires careful attention to leakage, missingness, repeated visits, and subject-level splitting. If visits from the same subject appear in both training and test sets, the model can learn subject-specific patterns and produce over-optimistic results. This makes data splitting one of the most important methodological choices in the project.

2.2 Machine Learning in Alzheimer’s Prediction

Traditional machine learning methods such as logistic regression, random forests, support vector machines, and gradient boosting have been widely used for Alzheimer classification and prognosis. These methods remain important because structured clinical variables are often highly predictive and easier to interpret than complex imaging features. XGBoost and CatBoost are particularly useful for tabular data because they can model nonlinear relationships, missingness patterns, and feature interactions.

2.3 Deep Learning and MRI-Based Classification

Deep learning has advanced Alzheimer prediction by allowing models to learn patterns from MRI images. CNN models can extract spatial features from brain images that may not be captured directly by tabular variables. Transfer learning, ResNet-style models, and custom CNN architectures have been used in prior MRI-based Alzheimer classification studies. However, image models are often more computationally expensive, require careful preprocessing, and may be sensitive to scanner differences, image quality, and class imbalance.

2.4 Multimodal Learning and Fusion

Multimodal learning attempts to improve prediction by combining different sources of information. In Alzheimer research, common modalities include MRI, PET, CSF biomarkers, cognitive tests, demographics, genetics, and clinical notes. Fusion can be performed early by concatenating features before prediction or late by combining predictions from separate modality-specific models. ALZ-Foresight uses late fusion in the final implemented branch by combining structured model outputs with CNN image outputs.

2.5 Explainability and Clinical Trust

Clinical decision-support systems require more than high accuracy. They must also provide interpretable and understandable outputs. Explainability techniques such as feature importance, SHAP, Grad-CAM, attention maps, and plain-language explanation panels help users understand why a model produced a specific prediction. In ALZ-Foresight, structured feature importance identifies influential ADNI variables, while the web application provides AI-generated explanation panels that describe the predicted class, relevant scores, and possible next steps in plain language.

2.6 Comparison with the Proposed Project

The proposed project differs from many isolated research experiments because it combines both model development and software implementation. It includes a leakage-safe structured branch, CNN image branch, multimodal late-fusion branch, secure web dashboard, patient record management, prediction API, visualization, explainability, logging, model versioning, and formal test validation. This makes ALZ-Foresight a complete research prototype rather than only a standalone notebook.

Table 2.1: Representative related work summary

Ref.	Model	Focus	Result	Limitation
[1]	Temporal Graph Network	Progression forecasting	AUC 0.80	Limited interpretability and OOD robustness.
[2]	CNN	MRI classification	High accuracy	Imaging-only setup.
[3]	XGBoost-Cox	Prognosis	C-index 0.92	ADNI bias and limited external validation.
[4]	Multimodal Transformer	Subtype discovery	AUROC 0.73	No external validation.
[5]	Dual-CNN	MRI classification	99.6% accuracy	MRI-only and computationally heavy.
[9]	GBM / RF / SVM / XGBoost	MCI conversion	AUC 0.875	Small sample and PET dependency.
[13]	XAI review	Explainability	Review findings	Need for clinically meaningful explanation.
[17]	DISFC	Long-term prediction	AUC 0.96	Multimodal complexity and small cohort.

Chapter 3

System Requirements Specification

3.1 Purpose

This chapter summarizes the system requirements for ALZ-Foresight. The SRS defines what the system must accomplish, who will use it, the required functions, non-functional attributes, interface requirements, constraints, and performance expectations. The intended audience includes the project supervisor, development team, evaluators, researchers, and potential clinical users.

3.2 Scope

The system supports research-oriented Alzheimer stage classification using multimodal ADNI data. It includes structured data ingestion, subject-level splitting, feature processing, CNN-based MRI classification, late-fusion prediction, web-based user management, patient management, prediction inference, visualization, explainability, audit logging, and model versioning.

The system does not provide certified medical diagnosis, regulatory approval, direct EHR integration, or clinical deployment. It is designed for academic research and educational demonstration.

3.3 User Characteristics

Table 3.1: Main user groups

User Group	Description
Clinician	Reviews patient profiles, enters assessment scores, requests predictions, and views explanations.
Researcher	Reproduces experiments, compares models, evaluates metrics, and studies feature behavior.
Admin	Manages users, roles, model versions, and audit logs.
Caregiver/Patient	Views simplified risk insights and recommendations when authorized.
Evaluator	Reviews documentation, system behavior, results, and project completeness.

3.4 Functional Requirements

Table 3.2: Functional requirements summary

Req. ID	Function	Description
FR-01.1	Data ingestion and cohorting	Import ADNI data, harmonize identifiers and visits, create leakage-safe subject-level splits, and produce reproducible datasets.
FR-01.2	Multimodal feature encoders	Convert structured variables and MRI data into usable model inputs through preprocessing and CNN-based feature learning.
FR-01.3	Multimodal fusion layer	Combine structured and image-based outputs using late fusion and support future early-fusion variants.
FR-01.4	Prediction heads	Produce current stage classification outputs and support future progression-aware risk horizons.
FR-01.5	Calibration and XAI artifacts	Support interpretable prediction through feature importance, probability outputs, and explanation panels.
FR-02.1	RBAC authentication gateway	Provide login, JWT authentication, and role-based access control.

Req. ID	Function	Description
FR-02.2	Clinical CRUD services	Manage patient records, assessments, medications, activities, and appointments.
FR-02.3	Inference proxy and health	Validate prediction requests, call the Python model layer, and return standardized prediction responses.
FR-03.1	Risk and longitudinal UI	Display patient risk, confidence, probability breakdowns, and longitudinal clinical context.
FR-03.2	XAI UI panels	Display explanation panels that describe the result, model reasoning, and patient-facing recommendations.
FR-04.1	Audit logger	Log sensitive actions such as login, prediction, update, delete, and error events.
FR-04.2	Model registry	Track active model files, versions, and deployment status.
FR-04.3	Health monitor	Support health checks, graceful errors, and timeout handling.
FR-04.4	Research exporter	Support anonymized exports for research reporting.
FR-05.1	Policy guardrails	Enforce research-only use and ADNI data-use restrictions.
FR-05.2	Risk education module	Explain risk, confidence, uncertainty, and prediction limitations in plain language.

3.5 Non-Functional Requirements

Table 3.3: Non-functional requirements summary

Req. ID	Attribute	Requirement Summary
NF-R1	Security	Protect users, patient records, prediction endpoints, and model artifacts through authentication, RBAC, input validation, and secure logging.

Req. ID	Attribute	Requirement Summary
NF-R2	Reliability	Maintain stable service behavior, handle downstream errors, and return graceful responses when AI insight generation or model subprocesses fail.
NF-R3	Maintainability	Separate frontend, backend, and ML service layers with clear interfaces and modular responsibilities.
NF-R4	Portability	Support local development and container-based deployment using environment variables and service separation.
NF-R5	Extensibility	Allow new model versions, new modalities, new fusion strategies, and additional explanation methods to be added later.
NF-R6	Accessibility and reusability	Use reusable UI components and make results readable for clinicians, researchers, and evaluators.
NF-R7	Resource utilization	Keep tabular prediction under three seconds and support image processing within practical limits.
NF-R8	UI serviceability	Provide clear UI feedback, error states, status indicators, and usable dashboard layouts.
NF-R9	User activity logging	Record clinically and security-relevant actions in an auditable way.

3.6 Interface Requirements

3.6.1 User Interfaces

The web interface includes login screens, patient dashboards, doctor dashboards, prediction forms, result panels, probability bars, explanation cards, appointment pages, activity pages, medication pages, reports, and messages. The UI is intended to support both clinician workflow and research demonstration.

3.6.2 Software Interfaces

The backend exposes REST APIs for authentication, patients, assessments, medications, activities, appointments, predictions, model information, audit logs, and health checks. The Python ML layer provides prediction execution through a script or model service. MongoDB Atlas stores users, patients, clinical records, predictions, audit logs, and model metadata.

Table 3.4: Key software interfaces

Interface	Purpose
/api/auth/login	Authenticates users and returns JWT tokens.
/api/auth/register	Registers users and returns access token.
/api/patients	Supports patient list, update, and deletion with RBAC.
/api/predict	Accepts clinical input values and returns model prediction.
/api/health	Confirms backend service status and timestamp.
Python model script	Loads active XGBoost model and performs inference.
MongoDB Atlas	Stores persistent application data.

3.7 Performance Requirements

The system targets fast prediction response for structured data. The tabular prediction requirement is three seconds or less. In testing, direct XGBoost inference completed in 649 ms, which satisfies this requirement. Full API response with GPT-4o AI insights took approximately 6.4 seconds, but the diagnostic result itself is generated before the insight-generation enrichment.

3.8 Design Constraints

The system is constrained by ADNI data-use requirements, the use of de-identified data, the need to prevent subject leakage, compute limitations related to MRI image processing, and the selected technology stack. The stack uses React, Node.js/Express, MongoDB Atlas, Python, XGBoost, CNN modeling, and AI-generated explanation support.

Chapter 4

System Design Document

4.1 Design Purpose

This chapter describes how ALZ-Foresight was designed to satisfy the requirements in the SRS. The design covers the architecture, components, data design, interfaces, user interface, and traceability between requirements and system modules.

4.2 Architecture Overview

ALZ-Foresight follows a three-tier client-server architecture. The frontend is a React web application. The backend is a Node.js/Express API that handles authentication, role-based access control, patient management, prediction orchestration, and audit logging. The machine learning layer is a Python-based service or script layer that loads trained model artifacts and returns prediction outputs. MongoDB Atlas stores persistent application data. Heavy artifacts such as MRI-related outputs or future Grad-CAM images can be stored as files or objects and referenced in the database.

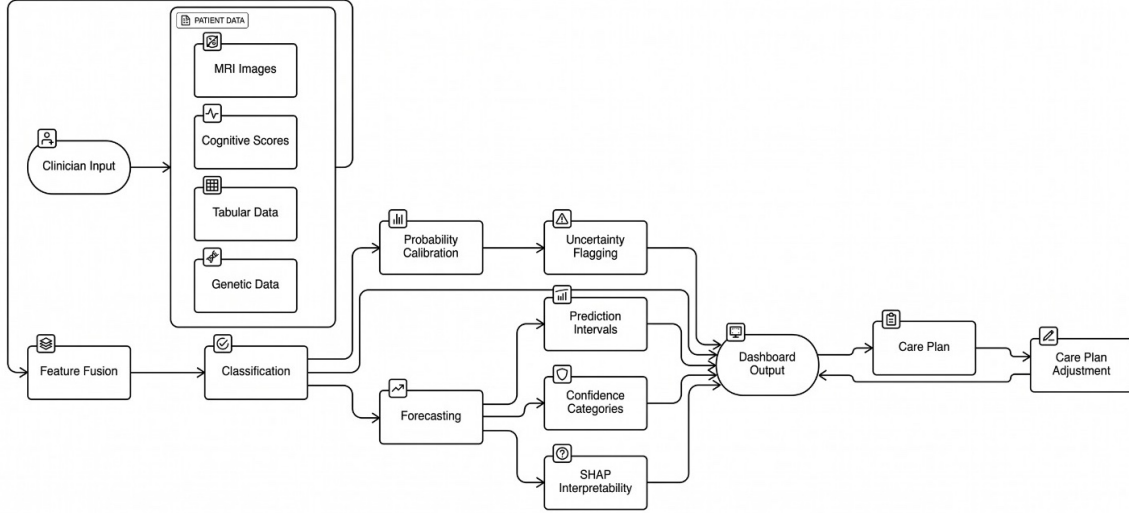


Figure 4.1: High-level ALZ-Foresight architecture showing multimodal input, classification, fusion, calibration, interpretability, and dashboard-based output.

4.3 Technology Stack

Table 4.1: Technology stack

Layer	Technologies
Frontend	React, routing, dashboard components, charting libraries, responsive UI.
Backend API	Node.js, Express, JWT authentication, RBAC middleware, validation, prediction routes.
Database	MongoDB Atlas with Mongoose ODM for users, patients, assessments, predictions, logs, and model metadata.
ML structured branch	Python, pandas, numpy, scikit-learn, XGBoost, CatBoost, imbalanced-learn.
ML image branch	CNN-based image classification for CN, MCI, and AD classes.
Fusion branch	Late fusion combining structured and CNN prediction outputs.
AI insights	GPT-4o-generated plain-language interpretation and recommendation panels.
Testing	Functional, security, reliability, inference-time, and endpoint testing.

4.4 Component Design

4.4.1 Frontend Components

The frontend contains authentication screens, role-aware navigation, patient dashboards, prediction forms, risk visualization panels, longitudinal score views, appointment views, medication views, activity tracking pages, and AI explanation panels. These components are designed to present complex prediction outputs in a simple interface.

4.4.2 Backend Components

The backend API includes the following modules:

- Authentication and RBAC module.
- Patient and clinical data management module.
- Prediction orchestration module.
- Input validation module.
- Audit logging module.
- Model versioning module.
- Health and reliability module.

4.4.3 Machine Learning Components

The machine learning side consists of:

- Structured ADNI preprocessing pipeline.
- XGBoost and baseline model training.
- Feature-importance ranking and Top-15 feature selection.
- CNN image classification branch.
- Late-fusion prediction branch.
- Evaluation and reporting scripts.

4.5 Data Design

4.5.1 ADNI Data Sources

The structured branch uses six ADNI tables. These tables are merged at visit level using RID and normalized visit codes.

Table 4.2: ADNI source tables

Table	Role
ADNIMERGE	Main visit-level anchor table.
Dementia_Rating	Dementia severity and CDR-related measures.
Diagnostic_Summary	Diagnostic outcomes and labels.
Cognitive_Scores	Cognitive assessment variables.
PTDEMOG	Demographic information.
ApoE_Genotyping	APOE-related genetic variables.

4.5.2 Application Database Collections

Table 4.3: MongoDB collections

Collection	Description
Users	Stores user accounts, roles, authentication information, and active status.
Patients	Stores patient demographic and clinical summary information.
Assessments	Stores cognitive and clinical visit-level measures.
Medications	Stores medication plans, dosage, adherence, and history.
Activities	Stores daily living and lifestyle records.
Appointments	Stores scheduled clinical appointments and status.
Predictions	Stores prediction result, probabilities, confidence, and model metadata.
ModelVersions	Stores active model identity, file reference, metrics, and version history.
AuditLogs	Stores authentication, CRUD, prediction, error, and system action logs.

4.6 User Interface Design

The UI was designed as a dark dashboard interface with role-aware views. Patients can see simplified information, while doctors can manage patients, run predictions, and view risk outputs. The prediction page accepts key clinical values such as CDRSB, MMSE

baseline, and ADAS13, then displays predicted diagnosis, confidence, class probabilities, and explanation cards.

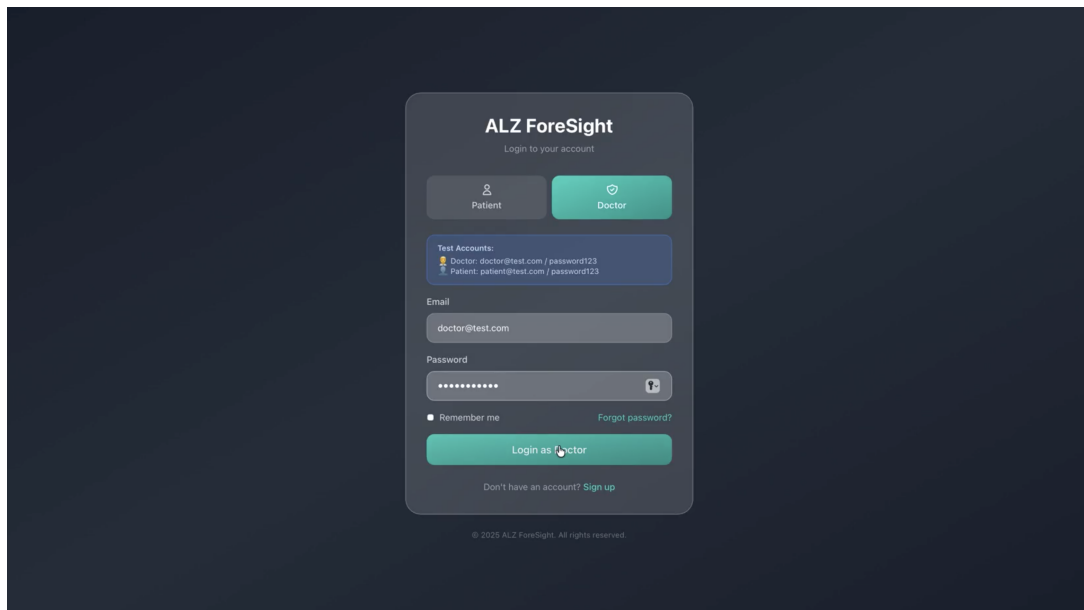


Figure 4.2: Example login interface for ALZ-Foresight.

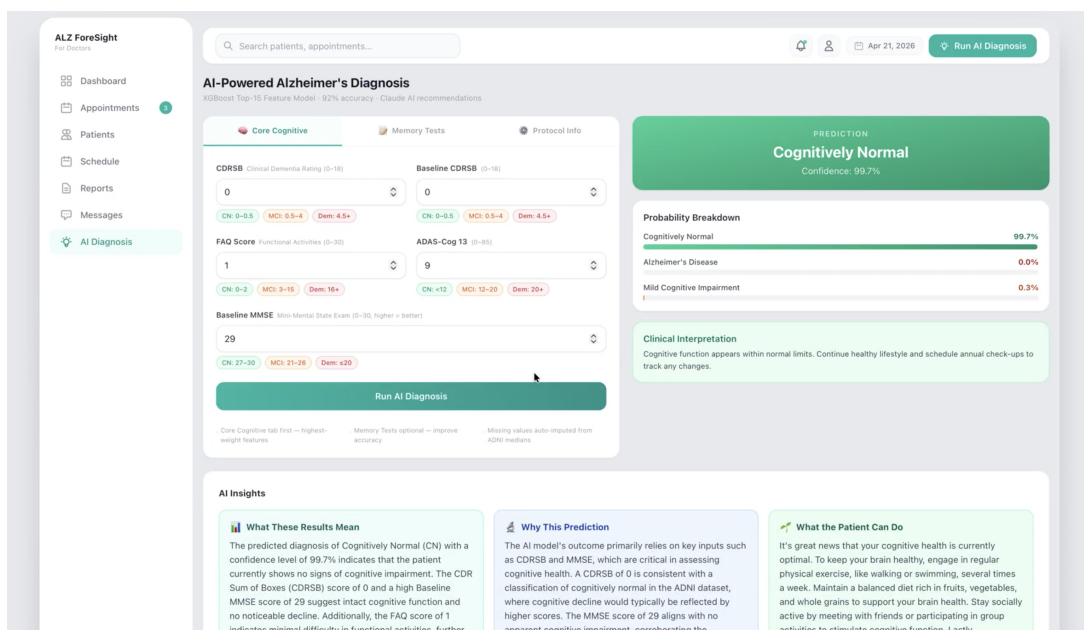


Figure 4.3: Example prediction interface showing input form and diagnosis result panel.

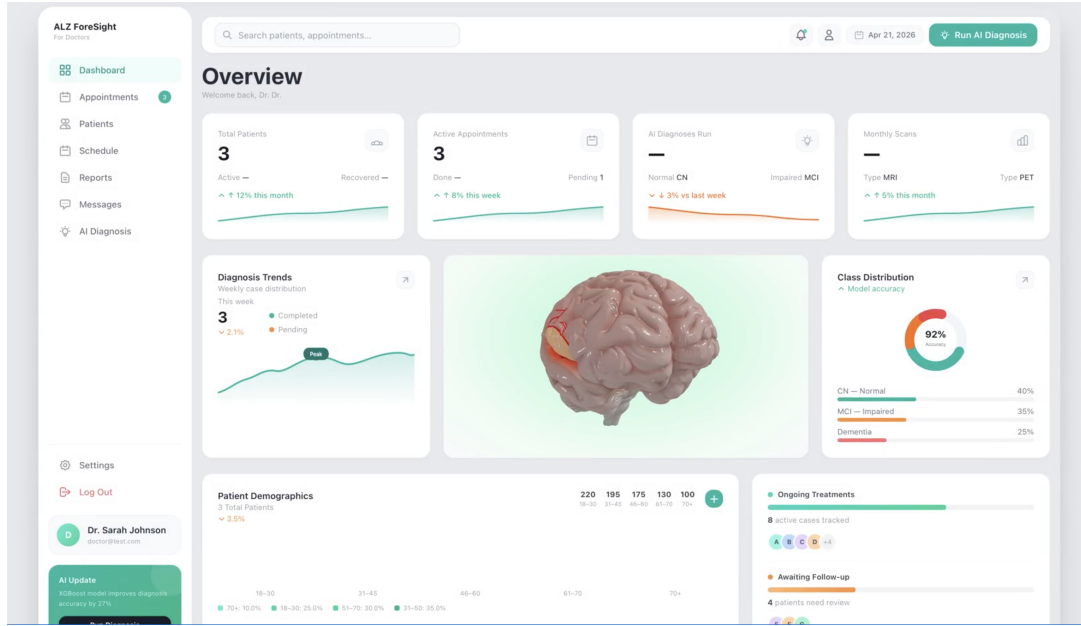


Figure 4.4: Example doctor dashboard for patient management and clinical workflow.

4.7 Requirements Traceability Matrix

Table 4.4: Requirements-to-design traceability matrix

Requirement	Design Component	Validation Method
FR-01.1	Structured ADNI ingestion and RID split pipeline	Data processing checks and model dataset shapes.
FR-01.2	Structured feature processing and CNN image branch	Model training outputs and hold-out evaluation.
FR-01.3	Late-fusion module	Fusion metrics and confusion matrix analysis.
FR-02.1	Auth and RBAC middleware	Login, invalid credentials, RBAC, and token tests.
FR-02.2	Patient CRUD services	Registration, update, and delete tests.
FR-02.3	Prediction endpoint and Python inference layer	Prediction API test and inference-time test.
FR-03.1	Risk visualization UI	Probability bars, diagnosis badge, and confidence display tests.
FR-03.2	Explanation panels	XAI visualization test.
FR-04.1	Audit logging	Audit creation and log integrity tests.

Requirement	Design Component	Validation Method
FR-04.2	Model versioning	Active model version and switching tests.
NF-R1	Security controls	JWT forgery, NoSQL injection, privilege escalation, and unauthenticated access tests.
NF-R3	Maintainable modular architecture	Separation of frontend, backend, database, and ML code.
NF-R7	Resource utilization	649 ms tabular inference result.

Chapter 5

System Implementation and Methodology

5.1 Implementation Overview

The implementation phase converted the project proposal, SRS, and SDD into a working prototype. The implementation combined machine learning experimentation with web application development. The ML side focused on ADNI data integration, structured preprocessing, XGBoost modeling, CNN image classification, late fusion, and evaluation. The application side focused on authentication, patient management, prediction API integration, visualization, explanation, logging, and model management.

5.2 Structured ADNI Pipeline

The structured pipeline merged multiple ADNI tables at visit level. RID was used as the subject identifier. The system performed train, validation, and test splitting by RID before applying downstream preprocessing. This prevented visits from the same subject from appearing across different partitions.

After identifiers and leakage-prone variables were excluded, missingness was computed on the training split only. Columns with training missingness greater than 0.45 were removed. This produced 111 processed features, including 88 numeric and 23 categorical variables.

Table 5.1: Feature processing summary

Item	Value
Missingness threshold	0.45
Dropped columns	139
Remaining processed features	111
Numeric features	88
Categorical features	23

5.3 Structured Model Development

The structured modeling phase evaluated XGBoost, CatBoost, and logistic regression. XGBoost achieved the strongest structured benchmark result. The updated structured branch then trained a 111-feature XGBoost model, extracted feature importance, selected the top 15 features, and applied hyperparameter tuning.

Table 5.2: Structured benchmark results

Model	Acc.	Prec.	Rec.	F1	ROC-AUC
XGBoost	0.9218	0.9224	0.9218	0.9219	0.9836
CatBoost	0.9109	0.9113	0.9109	0.9110	0.9824
Logistic Regression	0.7661	0.7780	0.7661	0.7663	—

5.4 Feature Importance and Top-15 Selection

Feature importance analysis showed that clinical severity and cognitive variables dominated prediction. The most influential variables included CDRSB, CDRSB baseline, FAQ, PTCOGBEG, mPACCtrailsB, ORIGPROT, LDELTOTAL baseline, mPACCdigit baseline, PTADDDX, MMSE baseline, and ADAS13.

Table 5.3: Top 15 selected features

Rank	Feature	Importance
1	CDRSB	0.1997
2	CDRSB_bl	0.0818
3	FAQ	0.0480
4	PTCOGBEG	0.0367
5	mPACCtrailsB	0.0270
6	ORIGPROT	0.0233
7	LDELTOTAL_BL	0.0209
8	mPACCdigit_bl	0.0198
9	PTADDX	0.0143
10	mPACCdigit	0.0140
11	MMSE_bl	0.0136
12	mPACCtrailsB_bl	0.0122
13	FSVERSION_bl	0.0096
14	VISDATE_ptd	0.0094
15	ADAS13	0.0092

Some of these variables, such as CDRSB and MMSE, are closely related to diagnostic criteria. For this reason, future work should evaluate model performance after removing proxy variables to measure how strongly the model depends on clinically direct indicators.

5.5 CNN Image Branch

The CNN image branch was implemented to process MRI samples corresponding to CN, MCI, and AD classes. The CNN produced class probability outputs, which were later used for multimodal late fusion. The MRI examples used in the report show representative class images from left to right: CN, MCI, and AD.

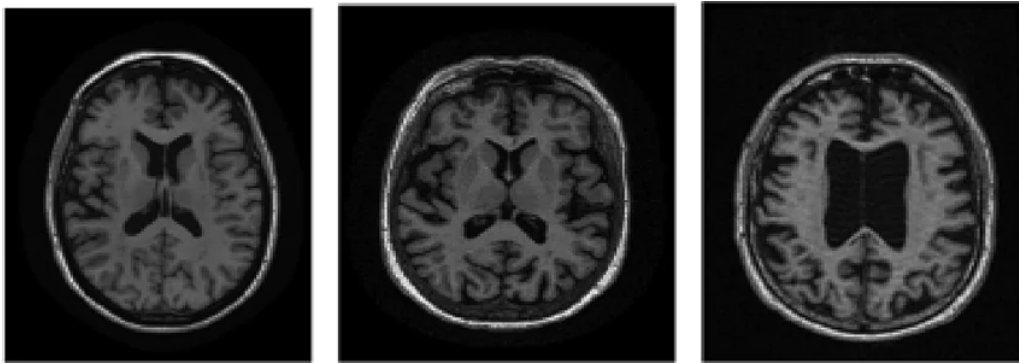


Figure 5.1: Representative MRI samples corresponding to cognitively normal, mild cognitive impairment, and Alzheimer’s disease, shown from left to right.

5.6 Late-Fusion Multimodal Branch

The late-fusion branch combined structured model outputs with CNN image model outputs. In the final fusion experiment, the CNN branch was assigned a weight of 0.75. The fused model was evaluated on a holdout set of 745 samples. This confirms that the final project implementation is multimodal in practice, not only in conceptual architecture.

5.7 Web Application Implementation

The web application includes:

- User login and registration.
- JWT-based authentication.
- Role-based access control.
- Patient registration, update, and deletion.
- Prediction form and result panel.
- Probability breakdowns for CN, MCI, and Dementia.
- AI-generated recommendation panels.
- Longitudinal score display.
- Audit logging.
- Model versioning and switching support.

The backend uses Node.js/Express. The database uses MongoDB Atlas. Model inference is performed through a Python layer that loads the active XGBoost model file. GPT-4o is used to generate structured AI insight panels after prediction.

5.8 Deployment Environment

The tested environment used local development with a Node.js/Express backend on port 5001, MongoDB Atlas, the Python XGBoost model file `xgb_model_top15.joblib`, and GPT-4o for AI insights. The SRS also defines a Vercel-based setup with React frontend, backend API, ML service, and MongoDB.

Chapter 6

Testing and Evaluation

6.1 Testing Strategy

Testing was designed to verify that the system works correctly from both software and machine learning perspectives. The tests covered authentication, security, patient management, input validation, prediction inference, timeout handling, visualization, explainability, audit logging, model management, inference time, and reliability.

6.2 Test Case Summary

Table 6.1: System test case summary

Test ID	Test Case	Requirement	Result
TC-06	Login returns JWT token	FR-02.1	PASS
TC-07	Invalid credentials rejected	FR-02.1	PASS
TC-08	Role-based access control	FR-02.2	PASS
TC-09	Patient registration	FR-03.1	PASS
TC-10	Patient record update	FR-03.2	PASS
TC-11	Patient record delete	FR-03.3	PASS
TC-12	Input validation	FR-04.1	PASS
TC-13	Prediction inference	FR-05.1	PASS
TC-14	API timeout handling	FR-05.2	PASS
TC-15	Risk visualization	FR-06.1	PASS
TC-16	Longitudinal data view	FR-06.2	PASS
TC-17	XAI visualization	FR-06.3	PASS
TC-18	Audit log creation	FR-07.1	PASS
TC-19	Log integrity	FR-07.2	PASS
TC-20	Model versioning	FR-08.1	PASS
TC-21	Model switching	FR-08.2	PASS
TC-22	Inference time	NFR-01	PASS
TC-23	Security testing	NFR-02	PASS
TC-24	System reliability	NFR-03	PASS

6.3 Functional Test Results

All functional tests passed. The login endpoint returned a valid JWT token. Invalid credentials were rejected with a generic error message. RBAC prevented patients from accessing doctor-only endpoints. Patient registration, update, and deletion workflows were confirmed. Prediction input validation rejected empty or malformed requests while allowing optional fields to be imputed by the preprocessing pipeline.

6.4 Prediction API Evaluation

The prediction endpoint returned diagnosis label, confidence score, and class probability distribution. In the tested example, the input contained $\text{CDRSB} = 2.0$, $\text{MMSE baseline} = 24$, and $\text{ADAS13} = 18$. The model predicted MCI with confidence 0.7928 and returned probabilities for CN, Dementia, and MCI. Direct XGBoost inference took 649 ms, satisfying the tabular prediction target of three seconds. The full API response with AI insights took approximately 6.4 seconds.

6.5 Visualization and Explainability Evaluation

The UI displayed a color-coded diagnosis badge, confidence percentage, and probability bars. The doctor dashboard displayed important patient scores such as CDRSB, MMSE, ADAS-Cog 13, and FAQ. Explainability was implemented through three AI-generated explanation panels:

1. What These Results Mean.
2. Why This Prediction.
3. What the Patient Can Do.

These panels explain the predicted diagnosis, identify influential input values, and provide plain-language recommendations.

6.6 Security Evaluation

Security tests confirmed that the system rejected forged JWT tokens, NoSQL injection attempts, privilege escalation attempts, and unauthenticated access. Patient-role tokens could not access doctor-only endpoints. Missing tokens produced HTTP 401 responses, while unauthorized role access produced HTTP 403 responses.

6.7 Reliability Evaluation

Reliability testing showed that five concurrent requests completed successfully without crashes, unhandled promise rejections, or port conflicts. The system also handled external AI insight failure gracefully by returning the core prediction result even if GPT-based explanation generation failed.

Chapter 7

Results and Conclusions

7.1 Structured Model Results

The structured branch produced the strongest overall predictive performance. The earlier XGBoost benchmark achieved 92.18% test accuracy and 0.9836 ROC-AUC. The updated 111-feature XGBoost model achieved 90.92% test accuracy and 0.9088 weighted F1. The reduced Top-15 model improved generalization, and the tuned Top-15 model achieved 91.90% test accuracy, 0.9189 weighted F1, 0.9181 macro F1, 0.9193 balanced accuracy, 0.8735 MCC, 0.2642 log loss, and 0.9783 weighted one-vs-rest ROC-AUC.

Table 7.1: Structured branch model comparison

Model	Features	Val. Acc.	Test Acc.	Test F1
111-feature XGBoost	111	0.8827	0.9092	0.9088
Top-15 XGBoost	15	0.8891	0.9155	0.9154
Tuned Top-15 XGBoost	15	0.8827	0.9190	0.9189

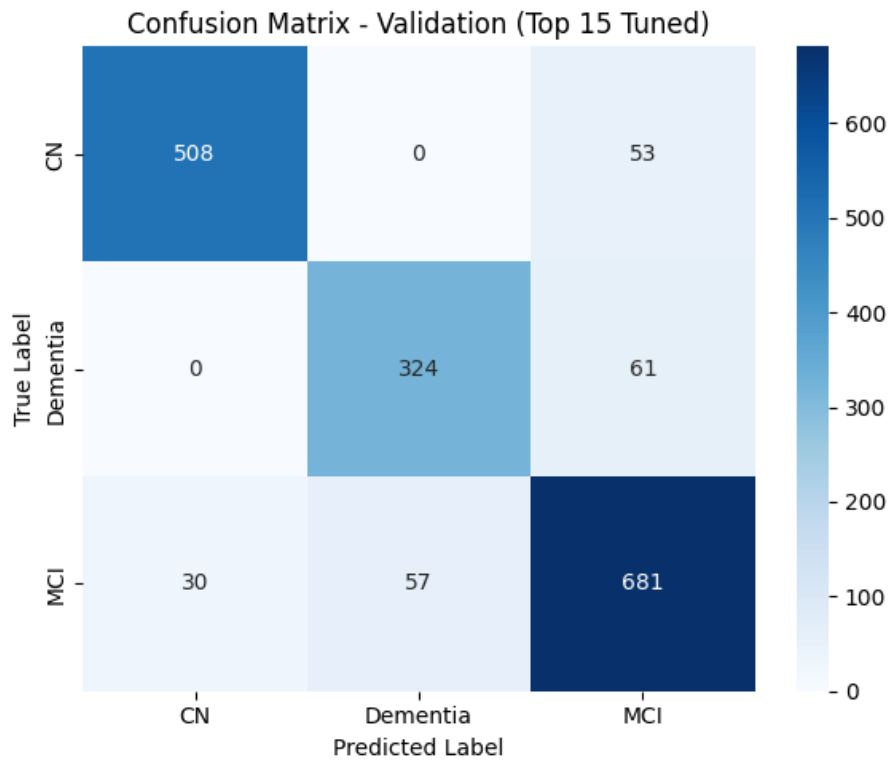


Figure 7.1: Validation confusion matrix for the tuned Top-15 XGBoost model.

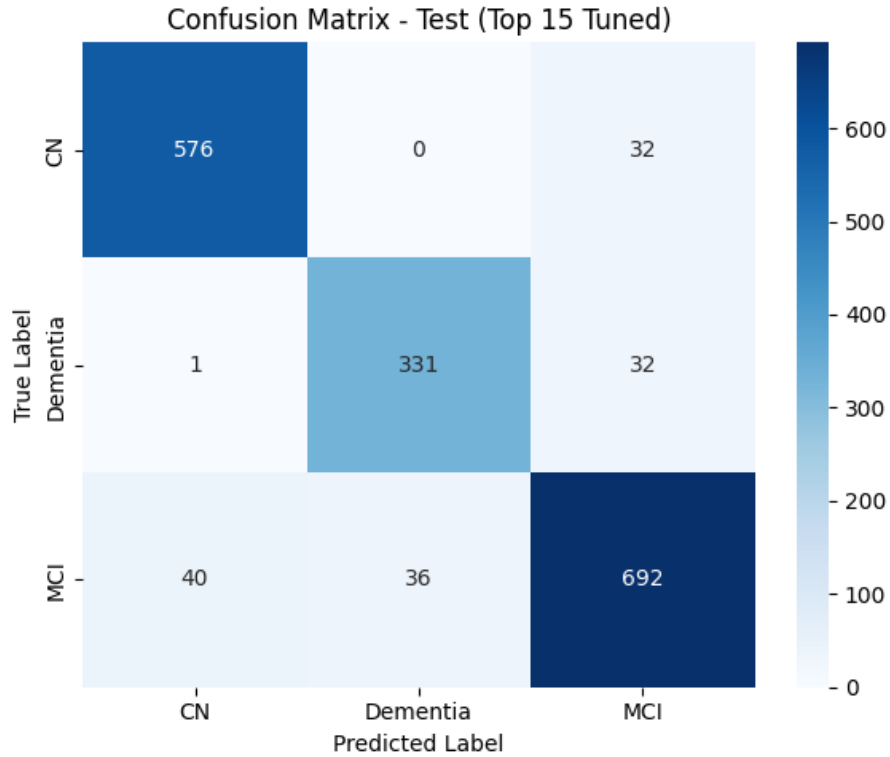


Figure 7.2: Test confusion matrix for the tuned Top-15 XGBoost model.

7.2 CNN and Fusion Results

The CNN image branch achieved 70.47% holdout accuracy, 0.7046 weighted F1, 0.6869 macro F1, and 0.8506 ROC-AUC. The late-fusion model achieved 73.15% accuracy, 0.7305 weighted F1, 0.6787 macro F1, and 0.8254 ROC-AUC. The fusion model improved accuracy compared with the CNN-only branch, but its ROC-AUC was lower than the CNN-only ROC-AUC, showing that the fusion method still requires further calibration and class-balancing refinement.

Table 7.2: Structured, CNN, and fusion performance

Branch	Accuracy	Weighted F1	Macro F1	ROC-AUC
Tuned Top-15 XGBoost	0.9190	0.9189	0.9181	0.9783
CNN image branch	0.7047	0.7046	0.6869	0.8506
Late-fusion model	0.7315	0.7305	0.6787	0.8254

Table 7.3: Class-level CNN and fusion performance

Branch	Class	Precision	Recall	F1
CNN	CN	0.7157	0.7157	0.7157
CNN	MCI	0.7127	0.7167	0.7147
CNN	AD	0.6374	0.6237	0.6304
Fusion	CN	0.7853	0.8930	0.8357
Fusion	MCI	0.7747	0.6431	0.7028
Fusion	AD	0.4554	0.5484	0.4976

7.3 System Validation Results

The system passed all 19 test cases. This means that the application met the tested requirements for authentication, RBAC, patient management, prediction, input validation, risk visualization, longitudinal data display, explanation, audit logging, model versioning, security, and reliability.

Table 7.4: System validation category summary

Category	Tests Passed	Result
Authentication and Security	3 / 3	PASS
Patient Management	3 / 3	PASS
Input Validation	1 / 1	PASS
AI Prediction	2 / 2	PASS
Visualization and Explainability	3 / 3	PASS
Audit and Logging	2 / 2	PASS
Model Management	2 / 2	PASS
Non-Functional Requirements	3 / 3	PASS
Total	19 / 19	PASS

7.4 Discussion

The project achieved its main goal of producing a working multimodal Alzheimer stage-classification system with a deployed web application. The structured XGBoost branch remained the strongest predictive component, confirming that tabular ADNI variables are highly informative. The CNN branch provided image-based classification capability, while late fusion showed that image outputs can be integrated with structured predictions.

The results also show that multimodal fusion is not automatically superior to the strongest individual branch. Although fusion improved accuracy over the CNN-only model, the structured model remained significantly stronger. This suggests that future work should investigate stronger CNN architectures, better MRI preprocessing, class weighting, probability calibration, and alternative fusion strategies.

The system validation results are especially important because they show that ALZ-Foresight is not only a model experiment. It is also a working application with authentication, patient management, prediction, visualization, explainability, logging, model versioning, and security behavior.

7.5 Conclusion

ALZ-Foresight successfully demonstrates a full project lifecycle from proposal to final evaluation. The project includes requirements, design, implementation, testing, and results. The final system integrates structured ADNI data, MRI-based CNN classification, late-fusion multimodal prediction, and a web application for clinical-style decision support.

The strongest model was the tuned Top-15 XGBoost model, which achieved 91.90%

test accuracy and 0.9783 weighted one-vs-rest ROC-AUC. The CNN branch achieved 70.47% accuracy, and late fusion achieved 73.15% accuracy. The system also passed 19/19 validation tests. These results show that the project produced both a reproducible machine learning pipeline and an operational web system.

7.6 Future Work

Future work should focus on improving the MRI pipeline, training stronger CNN or transformer-based image models, testing additional early and late fusion strategies, calibrating fusion outputs, evaluating proxy-feature ablation, adding more robust uncertainty estimation, validating the system outside the current ADNI configuration, and extending the system from static stage classification toward explicit longitudinal progression prediction.

Chapter 8

Lessons Learned and Problems Faced

8.1 Lessons Learned

Several lessons were learned during the project.

First, data leakage prevention is essential in longitudinal medical datasets. Subject-level splitting must be performed before preprocessing because row-wise splitting can produce misleadingly high model performance.

Second, structured clinical variables can be extremely predictive. Features such as CDRSB, FAQ, MMSE baseline, ADAS13, and mPACC-related measures carried strong signal. This demonstrated that strong tabular baselines are necessary before relying on deep imaging models.

Third, multimodal fusion requires careful calibration. Combining modalities does not guarantee automatic improvement over the strongest individual model. Fusion depends on the quality, balance, and calibration of each branch.

Fourth, implementation quality is as important as model quality. A prediction system must also handle login, validation, security, errors, logging, versioning, visualization, and reliability.

Fifth, explainability improves usability. Plain-language explanations and probability breakdowns help users understand model predictions more clearly than raw class labels alone.

8.2 Problems Faced

8.2.1 Handling Massive ADNI Data Files and Images

One of the main challenges was handling large ADNI files and MRI images. ADNI contains many tables, longitudinal visits, and imaging-related files. Managing these files required careful organization, repeated preprocessing, storage planning, and filtering. MRI files also introduced compute and memory challenges because image processing and CNN training are heavier than tabular modeling. The team had to work with selected representative image samples, manage file paths carefully, and avoid unnecessary duplication.

8.2.2 Missing Data and Table Integration

ADNI tables contain many missing values and inconsistent availability across subjects and visits. Merging multiple tables required RID and visit-code alignment. High-missingness columns had to be removed using training-only statistics. This reduced feature space while preserving useful clinical information.

8.2.3 Data Leakage Risk

The dataset contains multiple visits per subject. Without subject-level splitting, visits from the same subject could appear in training and testing partitions. This would leak subject-specific information. The project solved this by splitting by RID before transformations.

8.2.4 Class Imbalance

The class distribution was not perfectly balanced. MCI often formed the largest class, while AD had fewer samples in some image and fusion evaluations. This affected class-level performance, especially AD precision in the fusion model.

8.2.5 Proxy Variables

Some strong features, such as CDRSB and MMSE, are closely related to diagnostic criteria. This improves performance but raises a methodological question about whether the model is learning clinically direct proxies. Future work should evaluate performance after excluding such variables.

8.2.6 CNN and Fusion Limitations

The CNN branch produced meaningful but weaker results than the structured branch. The fusion branch improved accuracy over CNN alone, but AD precision remained limited and fusion ROC-AUC was lower than CNN-only ROC-AUC. This indicates that the fusion approach requires calibration, stronger image preprocessing, and more advanced training.

8.2.7 System Integration

Connecting the React frontend, Node.js backend, MongoDB Atlas database, Python model script, and AI insights introduced integration challenges. These included CORS, environment configuration, endpoint consistency, input validation, model file paths, and error handling.

8.2.8 Security and Reliability

The application needed to reject invalid credentials, forged tokens, unauthorized access, NoSQL injection attempts, and missing authentication headers. The team implemented JWT authentication, RBAC middleware, validation, and error handling to address these issues.

Appendix A

Acronyms

Acronym	Meaning
AD	Alzheimer’s Disease
ADNI	Alzheimer’s Disease Neuroimaging Initiative
ADAS-Cog	Alzheimer’s Disease Assessment Scale – Cognitive Subscale
APOE	Apolipoprotein E
AUC	Area Under the Curve
CDRSB	Clinical Dementia Rating – Sum of Boxes
CNN	Convolutional Neural Network
CN	Cognitively Normal
CRUD	Create, Read, Update, Delete
FAQ	Functional Activities Questionnaire
JWT	JSON Web Token
MCI	Mild Cognitive Impairment
ML	Machine Learning
MMSE	Mini-Mental State Examination
MRI	Magnetic Resonance Imaging
RBAC	Role-Based Access Control
ROC-AUC	Receiver Operating Characteristic Area Under the Curve
SRS	Software Requirements Specification
SDD	Software Design Document
XAI	Explainable Artificial Intelligence

Appendix B

Installation and Deployment Guide

B.1 Vercel Deployment Setup

The recommended deployment setup uses Vercel for hosting the ALZ-Foresight web application frontend. The GitHub repository is organized with the web application inside the `web-app` directory. Therefore, when importing the project into Vercel, the root directory should be set to:

`web-app`

The project can be cloned locally using:

```
git clone https://github.com/sezm00/ADNI-MULTIMODAL.git
cd ADNI-MULTIMODAL/web-app
npm install
npm run build
```

In the Vercel dashboard, the deployment settings should be configured as follows:

Table B.1: Vercel deployment settings

Setting	Value
Root Directory	<code>web-app</code>
Framework Preset	Vite / React
Install Command	<code>npm install</code>
Build Command	<code>npm run build</code>
Output Directory	<code>dist</code>

The frontend must be connected to the deployed backend API through an environment variable. In Vercel, the following environment variable should be added:

```
VITE_API_BASE=https://<backend-domain>/api
```

The backend folder exists inside the project under `web-app/backend`. The backend API, MongoDB Atlas database, and Python machine learning inference layer must be deployed or hosted separately from the Vercel frontend. Once the frontend is connected to the correct backend URL, Vercel automatically builds and redeploys the application whenever new commits are pushed to the connected GitHub branch.

B.2 Environment Variables

```
frontend/.env
```

```
VITE_API_BASE=http://localhost:4000/api
```

```
backend/.env
```

```
PORT=4000
```

```
MONGO_URI=mongodb://mongo:27017/adni
```

```
JWT_SECRET=change_me_dev
```

```
ML_API_URL=http://ml:5000
```

```
ml/.env
```

```
PORT=5000
```

```
MODEL_DIR=/models
```

```
ALLOW_CPU_FALLBACK=true
```

B.3 Health Checks

```
GET http://localhost:4000/api/health
```

```
GET http://localhost:5000/health
```

B.4 Smoke Test

The recommended smoke test is to log in, open a demo patient, add an assessment, request a prediction, view risk and explanations, and export the result.

Appendix C

ADNI Data Use Statement

This project uses ADNI data for research and educational purposes only. No attempt is made to re-identify subjects. No raw ADNI data are redistributed through this report. Subject identifiers are treated as pseudonymous research identifiers. The system is not a medical device and is not intended for direct clinical diagnosis or treatment decisions.

Bibliography

- [1] K. H. Aqil, P. Dumpuri, K. Ram, and M. Sivaprakasam, “Predictive modeling of Alzheimer’s disease progression: Integrating temporal clinical factors and outcomes in time series forecasting,” *Intelligence-Based Medicine*, vol. 10, p. 100159, 2024. doi: 10.1016/j.ibmed.2024.100159.
- [2] V. V. Chamundeeswari *et al.*, “Brain MRI analysis using CNN-based feature extraction and machine learning techniques to diagnose Alzheimer’s disease,” in *Proc. 2024 Conf.*, pp. 526–532, 2024. doi: 10.1109/icdcc62744.2024.10961923.
- [3] L. De Palma *et al.*, “A machine learning pipeline for prognostic modeling of Alzheimer’s disease using multimodal data,” *Sensors*, vol. 26, no. 5, p. 1523, 2026. doi: 10.3390/s26051523.
- [4] D. Machado Reyes *et al.*, “Identifying progression-specific Alzheimer’s subtypes using multimodal transformer,” *Journal of Personalized Medicine*, vol. 14, no. 4, p. 421, 2024. doi: 10.3390/jpm14040421.
- [5] A. M. El-Assy *et al.*, “A novel CNN architecture for accurate early detection and classification of Alzheimer’s disease using MRI data,” *Scientific Reports*, vol. 14, no. 1, 2024. doi: 10.1038/s41598-024-53733-6.
- [6] G. P. Shukla *et al.*, “Diagnosis and detection of Alzheimer’s disease using learning algorithms,” *Big Data Mining and Analytics*, vol. 6, no. 4, pp. 504–512, 2023. doi: 10.26599/bdma.2022.9020049.
- [7] M. Gupta *et al.*, “Adversarial network-based classification for Alzheimer’s disease using multimodal brain images,” *IEEE Access*, 2024. doi: 10.1109/access.2024.3381956.
- [8] B. K. Karaman *et al.*, “A deep survival model for predicting Alzheimer’s diagnosis based on multi-modal longitudinal data,” *Big Data Mining and Analytics*, vol. 9, no. 2, pp. 465–480, 2026. doi: 10.26599/bdma.2025.9020064.

- [9] M.-W. Lee *et al.*, “A multimodal machine learning model for predicting dementia conversion in Alzheimer’s disease,” *Scientific Reports*, vol. 14, no. 1, 2024. doi: 10.1038/s41598-024-60134-2.
- [10] M. Odusami *et al.*, “Machine learning with multimodal neuroimaging data to classify stages of Alzheimer’s disease: a systematic review and meta-analysis,” *Cognitive Neurodynamics*, 2023. doi: 10.1007/s11571-023-09993-5.
- [11] P. K. Pandey *et al.*, “Enhancing healthcare recommendation: Transfer learning in deep CNNs for Alzheimer disease detection,” *Frontiers in Medicine*, vol. 11, 2024. doi: 10.3389/fmed.2024.1445325.
- [12] S. U. Rehman *et al.*, “AI-based tool for early detection of Alzheimer’s disease,” *Heliyon*, vol. 10, no. 8, p. e29375, 2024. doi: 10.1016/j.heliyon.2024.e29375.
- [13] F. H. Saif *et al.*, “Explainable AI for Alzheimer detection: A review,” *Applied Sciences*, vol. 14, no. 22, p. 10121, 2024. doi: 10.3390/app142210121.
- [14] S. Muksimova *et al.*, “Multi-modal fusion and longitudinal analysis for Alzheimer’s disease classification,” *Diagnostics*, vol. 15, no. 6, p. 717, 2025. doi: 10.3390/diagnostics15060717.
- [15] S. K. Kim *et al.*, “Multimodal 3D deep learning for early diagnosis of Alzheimer’s disease,” *IEEE Access*, 2024. doi: 10.1109/access.2024.3381862.
- [16] C. Wang *et al.*, “A multimodal deep learning approach for prediction of cognitive decline,” *Translational Psychiatry*, vol. 14, no. 1, pp. 1–11, 2024. doi: 10.1038/s41398-024-02819-w.
- [17] Y. Wang *et al.*, “Predicting long-term progression of Alzheimer’s disease using multimodal deep learning,” *Journal of Translational Medicine*, vol. 22, no. 1, 2024. doi: 10.1186/s12967-024-05025-w.
- [18] S. Zarei *et al.*, “Statistical method for identification of Alzheimer’s disease in mild cognitive impairment with multimodal predictive markers,” *Basic and Clinical Neuroscience*, vol. 16, no. 2, 2025. doi: 10.32598/bcn.2024.2034.7.
- [19] S. Zhang *et al.*, “Machine learning on longitudinal multi-modal data enables the understanding and prognosis of Alzheimer’s disease progression,” *iScience*, vol. 27, no. 7, p. 110263, 2024. doi: 10.1016/j.isci.2024.110263.
- [20] R. E. Ali *et al.*, “Refined Alzheimer’s disease classification using MRI-based deep learning techniques,” in *2024 Intelligent Methods, Systems, and Applications*, pp. 569–574, 2024. doi: 10.1109/imsa61967.2024.10652832.